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FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

BTech-5th Semester

Report on

"Data Visualization on Expenses Dashboard"

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FACULTY OF ENGINEERING AND TECHNOLOGY(FET)

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CERTIFICATE

This is to certify that the Project entitled **"Data Visualization on Expenses Dashboard"** has been successfully presented by C Kumar Swamy , V Dhanush , K Ganesh Kumar Rao, Akash B, VK Ganesh bearing USN **KUB24CSE620, KUB23CSE152, KUB24CSE633, KUB24CSE605, KUB24CSE688,** students of 5th semester B. Tech for the partial fulfillment of the requirements for the **BTech in Computer Science & Engineering** of the **KISHKINDA UNIVERSITY** during the academic year 2025-2026

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Abstract

Expense management is an important part of personal and financial planning, but analyzing raw expense data from CSV files can be difficult and time-consuming. This project, **Data Visualization for Expenses Dashboard CSV Upload**, provides a simple and effective solution to this problem.

In this system, users upload a CSV file containing their expense details. The uploaded data is processed and displayed in the form of visual charts such as pie charts, bar graphs, and line charts. These visualizations help users easily understand their spending patterns, compare expenses across categories, and track changes over time.

The project reduces manual effort, improves clarity, and helps users make better financial decisions. It demonstrates how data visualization can transform raw data into meaningful insights in a simple and user-friendly way.

INTRODUCTION

In the modern digital era, effective financial management has become a necessity for individuals as well as organizations. Expenses Dashboard involves recording, monitoring, and analyzing financial expenditures over a period of time. However, raw numerical data stored in tables or spreadsheets is often difficult to understand and analyze.

Data Visualization plays a crucial role in simplifying financial data by representing it graphically through charts, graphs, and dashboards. This project focuses on developing a Data Visualization for Expenses Dashboard system, where users can upload a CSV (Comma-Separated Values) file containing expense data, and the system automatically generates meaningful visualizations.

By using CSV file uploads, the system eliminates manual data entry and allows users to analyze existing financial records quickly and efficiently.

OBJECTIVES OF THE PROJECT

The main objectives of this project are:

- To allow users to upload expense data in CSV format
- To process and analyze expense data automatically
- To visualize expenses using charts and graphs
- To provide category-wise, monthly, and trend-based analysis
- To help users understand spending patterns easily
- To support better budgeting and financial decision-making

PROBLEM STATEMENT

Most users store expense data in CSV or Excel files but face difficulties in analyzing the data effectively. The major problems include:

- Difficulty in interpreting large volumes of numerical data
- Lack of visual insights into spending behavior
- Time-consuming manual analysis
- No clear comparison between expense categories

Hence, there is a need for a system that can **take CSV input and convert raw expense data into visual representations** for easy understanding.

SCOPE OF THE PROJECT

The scope of the project includes:

- Uploading expense data through CSV files
- Automatic data processing and validation
- Category-wise expense visualization
- Monthly and yearly expense analysis
- Graphical representation using charts
- Personal expense analysis

The project can be extended in the future for business-level expense tracking and advanced analytics.

SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Hard Disk: 100 GB

Software Requirements

- Operating System: Windows / Linux
- Frontend: HTML, CSS, JavaScript
- Backend: Python
- Database (optional): MySQL
- Visualization Libraries: pandas / Matplotlib / Seaborn

DATASET DESCRIPTION (CSV FILE)

The system uses a CSV file as the input data source. Each row in the CSV file represents a single expense record.

Sample CSV Columns:

- Date
- Expense Category
- Amount
- Description

Ex: CSV File

```
Date,Category,Amount,Description
01-01-2025,Food,250,Lunch
02-01-2025,Transport,100,Bus Fare
03-01-2025,Shopping,1500,Clothes
```

The uploaded CSV file is read and processed by the system to generate visual insights.

METHODOLOGY

The project follows the steps given below:

Step 1: CSV File Upload

The user uploads a CSV file containing expense data using the application interface.

Step 2: Data Validation

The system checks:

- Correct file format
- Required columns
- Valid numerical values

Step 3: Data Processing

The data is processed to:

- Calculate total expenses
- Group expenses by category
- Aggregate monthly expenses

Step 4: Data Visualization

Charts and graphs are generated dynamically based on processed data.

Step 5: Analysis

Users analyze expense trends and identify high-spending categories.

DATA VISUALIZATION TECHNIQUES USED

The project uses the following visualization techniques:

- **Pie Chart**

  Displays category-wise expense distribution
  Helps identify which category consumes the most budget
- **Bar Chart**

  Compares expenses across months
  Useful for understanding monthly spending variations
- **Line Chart**

  Shows expense trends over time
  Helps analyze increase or decrease in expenses
- **Dashboard**

  Combines multiple charts
  Provides an overall financial overview

SYSTEM ARCHITECTURE

The system architecture consists of the following components:

- **User Interface** – Upload CSV file and view charts
- **Backend Module** – Reads and processes CSV data
- **Data Processing Module** – Performs calculations and grouping
- **Visualization Module** – Generates charts dynamically

Data flows from CSV upload to processing and finally to visualization.

MODULE DESCRIPTION

CSV Upload Module

- Accepts CSV file from user
- Validates file format and structure

Data Processing Module

- Reads CSV data
- Calculates totals and summaries
- Groups data by category and date

Visualization Module

- Generates pie, bar, and line charts
- Displays interactive graphs
- Updates visuals based on uploaded data

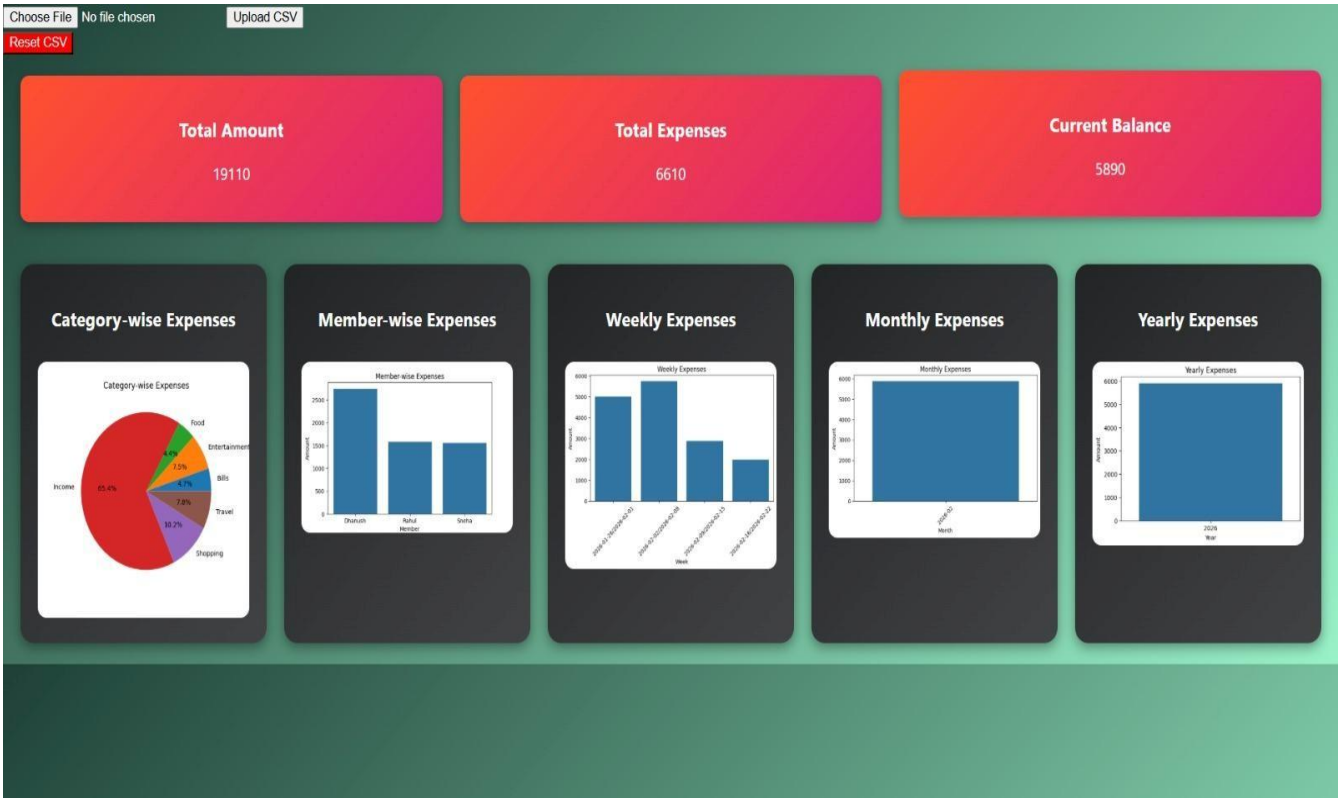
SAMPLE OUTPUT DESCRIPTION

After uploading the CSV file, the system displays:

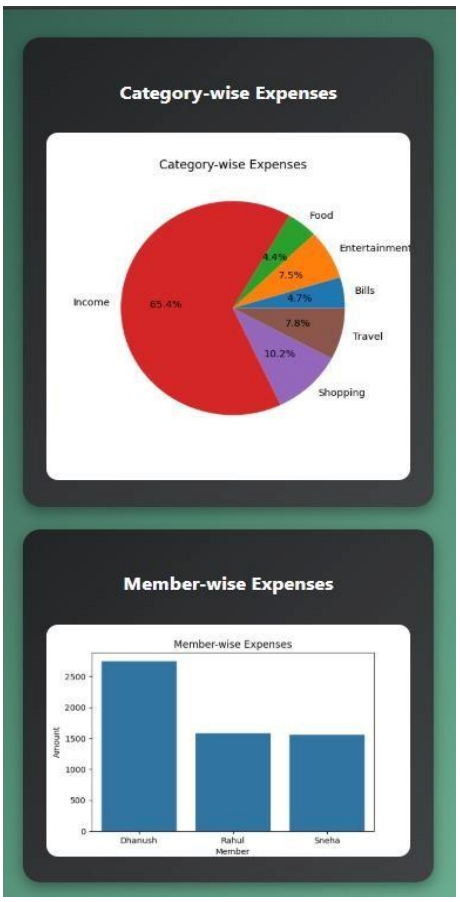
- Pie chart showing category-wise expense distribution
- Bar chart representing monthly expenses
- Line chart illustrating expense trends

These outputs help users quickly understand spending behavior without manually analyzing raw data.

Desktop/Laptop:



Mobile Phones:



ADVANTAGES OF THE SYSTEM

- No manual data entry required
- Easy analysis of existing expense data
- Saves time and effort
- Reduces calculation errors
- Improves financial awareness
- User-friendly interface

LIMITATIONS OF THE SYSTEM

- Depends on accuracy of CSV data
- Limited advanced analytics
- Cannot predict future expenses
- Requires structured CSV format

FUTURE ENHANCEMENTS

- Support for Excel file upload
- AI-based expense prediction
- Budget limit alerts
- Mobile application support
- Bank and UPI integration
- Cloud-based storage

CONCLUSION

The **Data Visualization for Expenses Dashboard using CSV Upload** project provides an efficient way to analyze financial data visually. By allowing users to upload CSV files and automatically generate charts, the system simplifies expense analysis and enhances financial decision-making. This project demonstrates the importance of data visualization in managing and interpreting real-world financial data.